



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 4E: Testing the 7-Segment Display

7-Segment Simulation

- Purpose of this lecture:
 - Show an example of a testbench to simulate an external FPGA interface
 - Provide an example of a testbench entity for re-use throughout the course

Testing of a 7-Segment Interface

- 7-segment interface:
 - Cathodes: 8 bit interface for digit
 - Anodes: 8 bit interface for which display to 'light' up
 - Display[0...7]: 8 display characters each 3 bits in length
- How to 'visualize' data
 - Use registers for storage if 'long' enough duration is set

Module Creation Steps

- First step:
 - Decode the Cathodes
 - Reverse the operation as seen in Lab 1
- Second Step:
 - Store the decoded value in display register when anode is driven low

Entity of Seg7 Simulator

```
entity seg7_interface_sim is  
    Port ( cathodes : in STD_LOGIC_VECTOR (7 downto 0) ;  
          anodes   : in STD_LOGIC_VECTOR (7 downto 0) ;  
          display0 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display1 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display2 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display3 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display4 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display5 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display6 : out STD_LOGIC_VECTOR (3 downto 0) ;  
          display7 : out STD_LOGIC_VECTOR (3 downto 0) ) ;  
end seg7_interface_sim;
```

Architecture of Seg7 Simulator

```
--decoder
  with cathodes select
    digit_decode <= X"0" when "11000000",
                      X"1" when "11111001",
                      X"2" when "10100100",
                      X"3" when "10110000",
                      X"4" when "10011001",
                      X"5" when "10010010",
                      X"6" when "10000010",
                      X"7" when "11111000",
                      X"8" when "10000000",
                      X"9" when "10010000",
                      X"A" when "10001000",
                      X"B" when "10000011",
                      X"C" when "11000110",
                      X"D" when "10100001",
                      X"E" when "10000110",
                      X"F" when others;
```

Architecture of Seg7 Simulator Continued

```
--Capture decoded character for each anode low signal (LATCHES! Only use in simulation)
char0 <= digit_decode when anodes(0) = '0' else char0;
char1 <= digit_decode when anodes(1) = '0' else char1;
char2 <= digit_decode when anodes(2) = '0' else char2;
char3 <= digit_decode when anodes(3) = '0' else char3;
char4 <= digit_decode when anodes(4) = '0' else char4;
char5 <= digit_decode when anodes(5) = '0' else char5;
char6 <= digit_decode when anodes(6) = '0' else char6;
char7 <= digit_decode when anodes(7) = '0' else char7;

display0 <= char0;
display1 <= char1;
display2 <= char2;
display3 <= char3;
display4 <= char4;
display5 <= char5;
display6 <= char6;
display7 <= char7;
```